

Uncertainty and Unruhigkeit:

A Galois Connection

Companion paper to: Syntax, Geometry, and Measurement

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Epigraph

"Show and tell are a conjugate pair: an actor issues orders, and a universe demonstrates the response."

Abstract

We develop a non-ecological thermodynamics parallel to the non-ecological mechanics of [Paper 1]. Temperature is treated as a primitive scalar field T on a Level 3 spacetime, satisfying the field equation $\Box(T) + (1/3)RT = 0$, derived from the free-streaming Liouville equation on the geodesic flow without any particle content. The central result is that curvature and temperature are conjugate in the Galois connection sense: their level surfaces coincide, causal isolation boundaries are isothermal surfaces, and curvature singularities are temperature point sources. A covariant four-current conservation law follows automatically from the field equation and the Bianchi identities, completing the ecology-free field-theoretic picture. The Unruh effect is reinterpreted as the canonical Level 1 thermodynamic result, requiring no quantum field theory. The equivalence principle is a tautology. The non-ecological Tolman relation $r_{\text{eff}} * T = 1/4\pi$ and the thermodynamic uncertainty relation $S * T^2 = 1/16\pi$ are shown to be the same equation. Fundamental constants are identified as self-consistency conditions at the Planck scale. The programme can be read as a partial response to Hilbert's sixth problem, approached from the informational rather than the mechanistic direction.

1. Introduction and Motivation

This paper is a companion to [Paper 1], which developed a unified foundational programme separating structural mathematics from ecological data across logic, geometry, and physics. Here we extract and develop the thermodynamic thread into a focused result: a non-ecological thermodynamics in which temperature is a primitive geometric scalar field, and curvature and temperature are conjugate in the Galois connection sense.

1.1 Hilbert's sixth problem

Hilbert's sixth problem, posed in 1900, calls for the axiomatic treatment of the physical sciences — specifically mechanics and probability — in the same spirit as the axiomatisation of geometry. It has never been fully resolved. The present programme can be read as a partial response, approached from the informational rather than the mechanistic direction: we provide axiomatic foundations for relativistic mechanics and thermodynamics, stripped of ecological assumptions, from a causal and geometric standpoint that Hilbert could not have anticipated but would surely have recognised. Carathéodory's 1909 axiomatisation of thermodynamics — which proceeds without Carnot cycles, ideal gases, or ecological substance, from the geometry of state space alone — is our most direct predecessor in spirit.

1.2 SR's contaminated origin

Einstein's 1905 paper was titled 'On the Electrodynamics of Moving Bodies' — its motivation was entirely electromagnetic. The postulate about the speed of light is at root a postulate about electromagnetic waves. The Level 0-1 framework of [Paper 1] achieves the necessary disentanglement, replacing light with the non-ecological maximum signal speed.

1.3 The Majumdar-Papapetrou warning

The Majumdar-Papapetrou solutions describe extremal black holes balanced by electromagnetic repulsion — deeply ecological, requiring charge from the standard model. We are chargeless by construction. These solutions are not available to us, and their elegance should not tempt us to reintroduce electromagnetism.

1.4 The equivalence principle as tautology

In our framework there is no mass and no force — only geodesic motion in a curved causal structure. The equivalence principle has no content to assert. Curvature is the deviation from straight causal propagation — acceleration and curvature are definitionally related. The principle dissolves into a tautology, which is not a failure but a success: we have found the right level of description.

1.5 Natural units as ecological bookkeeping

Our framework admits only c and π as non-ecological constants. The constants $\hbar$, k_B , G , and e are ecological bookkeeping — conversion factors between geometric and physical units. Their appearance in any formula signals ecological contamination. The Hawking temperature $T_H = \hbar c^3 / 8\pi G M k_B$ fails this diagnostic; its non-ecological residue $T_H = 1/4\pi r_s$ passes it.

2. Particle-Free Classical Thermodynamics

The central simplification of the non-ecological setting is the collapse of the adiabatic/isothermal distinction. In a particle-free setting, both collapse to a single geometric condition: **causal isolation**. Adibat = isotherm = causal isolation boundary. This is a

theorem, not an approximation. There are no spin-statistics complications: the temperature field is classical and geometric.

2.1 The four laws, recast

Zeroth. Causal contact implies equal boundary temperature.

First. Conservation from the Bianchi identities — no heat, no work in the ecological sense.

Second. Causal irreversibility — entropy non-decreasing along geodesics. Carathéodory's inaccessibility condition identified exactly with causal inaccessibility.

Third. $T = 0$ at causally flat regions where curvature vanishes.

2.2 The noise floor as event horizon

The noise floor — the informational boundary beneath which no structure is recoverable — is the event horizon expressed in thermodynamic language. They are the same boundary seen from the causal and thermodynamic sides of the adjunction respectively.

2.3 The non-ecological tensor inventory

Energy is an ecological form of entropy — the geometric entropy field weighted by ecological temperature and coupling constants. In our formalism energy, momentum, and force do not exist as primitives. The standard stress-energy tensor $T_{\mu\nu}$ is more honestly the **entropy-flux tensor** $\Sigma_{\mu\nu}$: it encodes the distribution of entropy flux and causal influence, without reference to energy or momentum. The Maxwell tensor $F_{\mu\nu}$ does not appear. We are chargeless by construction. Its absence is not an omission but a feature.

$$\begin{aligned} g_{\mu\nu} &= \text{causal structure tensor} \\ R_{\mu\nu\rho\sigma} &= \text{curvature tensor} \\ R_{\mu\nu} &= \text{geodesic focusing tensor} \\ G_{\mu\nu} &= \text{geometric divergence tensor} \\ \Sigma_{\mu\nu} &= \text{entropy-flux tensor [no } F_{\mu\nu}] \end{aligned}$$

3. The Level Hierarchy for Thermodynamics

Level 0 — Entropy as causal path counting

Entropy is the logarithm of the number of causally accessible microstates — the count of distinct causal paths. No dynamics, no Hamiltonian. The arrow of time follows immediately: causal order is irreversible.

Level 1 — Temperature from cone structure

The canonical Level 1 thermodynamic result is the Unruh effect, properly understood. An accelerating observer sits on a non-trivial isothermal surface. Their Rindler horizon is their personal noise floor. The temperature they perceive is the monopole coefficient of the temperature point source at their causal horizon: T proportional to a/c^2 . No particle physics. No $\hbar$. Acceleration is heating without chemistry.

Level 2 — Tolman-Ehrenfest

The first ecological level. The Minkowski metric enters, and with it the standard Tolman-Ehrenfest relation. SR thermodynamics as a special ecological case.

Level 3 — Temperature as a scalar field

At Level 3, temperature is simply a scalar field $T : M \rightarrow \mathbb{R}^+$ on the spacetime manifold — a geometric object as primitive as the

curvature tensor. Proceeding backward from Level 3 reveals what the special cases were concealing.

4. The Temperature Field Equation

4.1 From Raychaudhuri and Bianchi

Let isothermal surfaces $\Sigma_c = \{T = c\}$ be causal isolation boundaries. The congruence within Σ_c has vorticity $\omega_{\mu\nu} = 0$ (hypersurface orthogonality). At a non-expanding horizon the Raychaudhuri equation gives:

$$R_{\mu\nu} u^\mu u^\nu = -\sigma_{\mu\nu} \sigma^{\mu\nu} \leq 0$$

The contracted Bianchi identity gives:

$$\nabla^\mu R_{\mu\nu} = (1/2) \nabla_\nu R$$

Projecting onto the temperature gradient yields the integrability condition:

$$\nabla_\mu T \propto \nabla_\mu R$$

The gradients of T and R are aligned. Their level surfaces coincide. The temperature field and the Ricci scalar field foliate the manifold identically.

4.2 From the free-streaming Liouville equation

The non-ecological Boltzmann equation is the free-streaming Liouville equation on the geodesic flow: $u^\mu \nabla_\mu f = 0$. No particles, no collisions. Identifying T as the zeroth moment of the geodesic distribution, T proportional to $\rho = \int f d\Omega$, and commuting covariant derivatives — generating Ricci tensor terms via the Ricci identity — yields:

$$\square T + (1/3) R T = 0$$

The non-ecological temperature field equation. A wave equation with curvature-dependent effective mass $m^2 = R/3$. Contains only c and π . In flat spacetime $R = 0$ and $\Box(T) = 0$ — temperature is a free massless scalar. Compare the conformally coupled scalar $\Box(\phi) + (1/6)R\phi = 0$: our coefficient $1/3$ is twice the conformal value. We conjecture this non-invariance is the geometric expression of the second law.

4.3 The entropy field

With S proportional to $\log(T)$:

$$\square S + |\nabla S|^2 + (1/3)R = 0$$

A Hamilton-Jacobi equation — entropy is the action of the thermodynamic geometry. The analogy with the semiclassical limit of quantum gravity is striking and probably not accidental.

4.4 Covariant four-current conservation

The field equation $\Box(T) + (1/3)RT = 0$ admits a conserved covariant stress-energy tensor for the temperature field itself — an ecology-free conservation law following automatically from the field equation and the Bianchi identities, requiring no additional assumptions.

Define the stress-energy tensor of the temperature field:

$$\Theta^{\mu\nu} = \nabla^\mu T \nabla^\nu T - (1/2) g^{\mu\nu} (\nabla_\lambda T \nabla^\lambda T + (1/3)RT^2)$$

By virtue of the field equation $\Box(T) + (1/3)RT = 0$ and the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$, this tensor satisfies:

$$\nabla_\mu \Theta^{\mu\nu} = 0$$

This is the covariant conservation law for the temperature field — the non-ecological analogue of energy-momentum conservation, but now for the thermodynamic geometry itself. It is not imposed; it is derived. The geometry enforces the conservation without any ecological input, exactly as current conservation follows from the Maxwell equations in electromagnetism — though here there is no electromagnetism, no charge, and no particles. The conservation is purely geometric.

This completes the ecology-free field-theoretic picture. The temperature field T has a field equation, a conserved stress-energy tensor, and a Hamilton-Jacobi entropy field — all derived from the causal skeleton alone. The standard model is nowhere required.

5. Curvature and Temperature as a Conjugate Pair

5.1 The Galois connection

not side: causal isolation boundaries — level surfaces of T , freely determined by the causal order, requiring no ecological data.

neg side: curvature sources — singularities sourcing R , requiring ecological specification to particularise.

The Galois axioms follow from the alignment $\nabla_\mu T$ proportional to $\nabla_\mu R$: nothing is its own curvature source; the deep interior of an isolation boundary lies within it.

5.2 Curvature singularities as temperature point sources

Curvature singularities are distributional sources in $\Box(T) + (1/3)RT = 0$. In the exterior $R = 0$, so $\Box(T) = 0$ — the temperature field is sourced only by the singularity, exactly as the Newtonian potential satisfies $\nabla^2(\phi) = 0$ in the exterior with a point source at the origin. The Hawking temperature T_H is the monopole coefficient. Higher multipoles — dipole from rotation (Kerr), quadrupole from deformation — constitute a temperature multipole expansion not previously examined from this geometric angle.

5.3 The non-ecological Unruh effect

The Unruh effect follows from two geometric facts alone:

Fact 1. Level surfaces of T and R coincide — the boundary of a region of non-zero curvature is simultaneously an isothermal surface.

Fact 2. Curvature singularities are temperature point sources.

An accelerating observer is, by the equivalence principle-as-tautology, locally in a curved region. They perceive the monopole temperature of the effective curvature source at their Rindler horizon. No quantum fields. No Bogoliubov transformations. No $\hbar$. Unruhigkeit is conjugate to Unruheffekt: restlessness (the instruction) is conjugate to the thermal bath (the response). Their product is the geometric constant $1/4\pi$.

$$T = a / 2\pi c^2$$

5.4 The Schwarzschild temperature field

In the Schwarzschild exterior $R = 0$, so $\Box(T) = 0$. For static, spherically symmetric $T = T(r)$:

$$d/dr [(r^2 - r_s r) dT/dr] = 0$$

Integrating twice with $T \rightarrow 0$ as $r \rightarrow \infty$:

$$T(r) = (1/4\pi) \log(1 - r_s/r)$$

Monopole tail $T \sim -r_s/4\pi r$ at large r . Logarithmic divergence at $r = r_s$ — the horizon as extended temperature source. Hawking temperature:

$$T_H = 1 / 4\pi r_s$$

No $\hbar$, G , or k_B . Only π and the geometric length r_s .

5.5 The non-ecological Tolman relation

Temperature is proportional to the square root of the surface curvature of the isothermal surface, $\kappa = 1/r_s^2$. With $S = A/4 = \pi r_s^2$ and $r_{\text{eff}} = r_s$:

$$r_{\text{eff}} \cdot T = 1 / 4\pi$$

The non-ecological Tolman relation — temperature times effective radius is a pure geometric constant. Reduces to $T \sqrt{g_{00}} = \text{const}$ ecologically.

5.6 The thermodynamic uncertainty relation

From $S = \pi r_s^2$ and $T = 1/4\pi r_s$:

$$S \cdot T^2 = 1 / 16\pi$$

A product of conjugate quantities equalling a pure geometric constant. The Schwarzschild solution is the maximally coherent state — the equality is saturated. For Kerr (rotation, non-zero vorticity $\omega_{\mu\nu}$):

$$S \cdot T^2 \geq 1 / 16\pi$$

with equality only at zero rotation. This is the gravitational uncertainty principle. Most strikingly: the non-ecological Tolman relation and the thermodynamic uncertainty relation are **the same equation**. The uncertainty relation is already contained in the Tolman relation, properly understood in the non-ecological setting.

6. The Unruh Effect Reinterpreted [to be expanded]

Standard derivation via Bogoliubov transformations — ecological loading identified. Non-ecological restatement: T proportional to a/c^2 from Facts 1 and 2 of Section 5.3 alone. The Rindler horizon as Level 1 causal feature — the personal noise floor of the accelerating observer. The universe showing what is being attempted. Acceleration as heating without chemistry: the purest thermodynamic process.

7. The Equivalence Principle as Tautology [to be expanded]

No mass, no force, only geodesic deviation. Curvature is the deviation from straight causal propagation — acceleration and curvature are definitionally related. The dissolution/sharpening diagnostic: equivalence principle dissolves; Unruh effect sharpens. The tautology locates the correct level of description.

8. Uncertainty, Temperature, and the Full Conjugate Table [to be expanded]

Revisiting Paper 1's relativistic uncertainty: $x_B = Tc/2$, collapse structure, conjugate variables in a massless universe. The full conjugate table:

position / velocity (Paper 1, Level 1)
curvature / temperature (Paper 2, Level 3)

observation / instruction (logical level)

uncertainty / cost (Section 7 of Paper 1)

Vaccarino-Barnett: all four pairs are valid accounting numeraires. Kerr as next step: $S \cdot T^2 \geq 1/16\pi$ with angular momentum. The theorem of ecological uncertainty (Paper 1, Section 6.7): every Galois connection generates an uncertainty relation; every uncertainty relation is an instance of a Galois connection.

9. Synthesis and Open Questions [to be expanded]

Non-ecological mechanics and non-ecological thermodynamics as two faces of the causal skeleton. $\text{Box}(T) + (1/3)RT = 0$ as zeroth moment of the free-streaming Liouville equation — a derivation, not a postulate. The factor $1/3$ vs $1/6$: conjecture that non-invariance is the geometric content of the second law. Unruheigkeit is conjugate to Unruheffekt — the title states the theorem.

Open questions: (1) Kerr uncertainty relation — $S \cdot T^2 \geq 1/16\pi$ with angular momentum J . (2) Whether $\text{Box}(T) + (1/3)RT = 0$ follows from Bianchi identities alone without a Liouville derivation. (3) Hamilton-Jacobi structure of S and the semiclassical limit of quantum gravity. (4) Whether quantisation arises naturally at Level 3. (5) The Kerr temperature multipole expansion.

Connection to Noether's theorem [Paper 4, in preparation]: every continuous symmetry of the action generates a conserved quantity (Noether); every reciprocal coupling generates an uncertainty relation (our theorem). Noether counts what is free; our theorem prices what is not. The action read as dissipative cost of a path — the receipt for the trajectory.

Forward Reference: Paper 3

Vorticity, Helicity, and Curvature Rotons: Topological Thermodynamics of the Geodesic Flow. Vorticity $\omega = \text{curl}(v)$ corresponds to curvature $R_{\mu\nu\rho\sigma}$. Helicity $H = \int (v \cdot \omega) dV$ is the fluid analogue of gravitational entropy — a topological invariant counting vortex line linking, related to the Gauss linking integral and Stokes theorem exactly as the Bianchi identities relate local curvature to global causal structure. Kelvin's circulation theorem = the Bianchi identities.

Curvature rotons: the dispersion relation $\omega^2 = k^2 + R/3$ from the temperature field equation has a mass gap at $k=0$ of $\omega_{\text{min}} = \sqrt{R/3}$ — the roton signature. The characteristic scale is the Raychaudhuri focusing length $l_f \sim R^{-(1/2)}$. Curvature rotons may correspond to new oscillatory vacuum solutions of the Bianchi-constrained curvature dynamics — neither singular nor flat but periodic. In the ecological presentation: quasi-normal modes of the gravitational field derived from geometric first principles rather than perturbation theory.

Turbulence as thermodynamic equilibrium: the Kolmogorov cascade as the renormalisation group flow on the topological configuration space. The $k^{-(5/3)}$ spectrum as the fixed point of the thermodynamic uncertainty relation. Turbulence is the receipt for driving a fluid far from its geometric ground state.

10. Gauss, the Planck Length, and Fundamental Constants

We close by returning to the ecological diagnostic of Section 1, now equipped with the full thermodynamic framework. The fundamental constants are not primitive — they are fixed points of self-consistency conditions.

10.1 Gauss's theorem as uncertainty

An observer outside a horizon has access only to the surface integral — the flux of T and R through the boundary. The volume integral — the detailed internal distribution — is causally inaccessible. The uncertainty is the non-uniqueness of the internal distribution consistent with a given surface flux. Applied to our field equation:

$$\oint_{\partial V} T \, dA = \int_V \square T \, dV = -(1/3) \int_V RT \, dV$$

The left side is observable. The right side is not. The uncertainty relation $S \cdot T^2 = 1/16\pi$ is the statement that the Gauss flux through the isothermal surface is a geometric invariant. This is the holographic principle as a consequence of Gauss's theorem applied to the non-ecological temperature field.

10.2 The fundamental volume

The natural domain of integration is the ball $B(l_f)$ of radius $l_f \sim R^{-(1/2)}$ — the focusing length set by the Raychaudhuri equation. The Gauss uncertainty applied to this fundamental volume: you can observe the boundary flux of T through the surface of $B(l_f)$, but not the internal distribution of RT within it. The variance of the internal distribution gives the uncertainty in R conjugate to the observed T .

10.3 The Planck length as minimal event horizon

The self-consistency condition — singularity size equals its own focusing length — is:

$$r_s = l_f(r_s) = r_s / \sqrt{3}$$

In non-ecological units this fixed point coincides, when ecological units are reinstated, with the Planck length $l_P = \sqrt{\hbar G/c^3}$. The Planck length is the minimal event horizon — the smallest curvature singularity whose isothermal surface is self-consistent with the temperature field equation. It is not a quantum of space — it is a geometric lower bound.

10.4 Fundamental constants as self-consistency conditions

l_P – fixed point of $r_s = l_f(r_s)$: minimal event horizon

$\hbar$ – quantum of action: ecological unit of the syntax/semantics adjunction

k_B – conversion: geometric $T \sim 1/r_s$ to thermodynamic energy $k_B T$

G – coupling: $\Sigma_{\mu\nu}$ to $G_{\mu\nu}$, vanishing to Bianchi identities

e – absent: chargeless by construction

Each is the solution to a fixed-point equation at the boundary between the non-ecological skeleton and its ecological realisation. The fundamental constants are the Galois connection evaluated at those fixed points — the precise numerical expression of how much ecological data is needed to instantiate a particular world on the geometric skeleton. The ecological diagnostic of Section 1 is now complete: we opened by saying that $\hbar$, G , k_B , and e are bookkeeping; we close by saying what they are bookkeeping of. The universe is showing what was attempted when it chose these values. The ledger is the Planck scale.

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